

22PH202 & ELECTROMAGNETISM AND MODERN PHYSICS

UNIT I – ELECTRICITY

- Electric dipole
- polarization

Introduction

- The electric dipole moment p of two charges $+q$ and $-q$ separated by a distance l is a vector of magnitude $p = ql$ with a direction from the negative to the positive charge.
- The polarized molecules are attracted toward the charged object because the field increases in the direction of the charged object.

Dielectrics



- A dielectric is a **nonconducting** material, an electrical insulator, such as rubber, glass, or waxed paper.
- When a dielectric is inserted between the plates of a capacitor, the capacitance increases *by a dimensionless factor κ , which is called the **dielectric constant of the material**.*
- **The dielectric constant** is the ratio of the field without the dielectric (E_0) to the net field (E) with the dielectric:

$$\kappa = E_0 / E$$

• Dielectrics

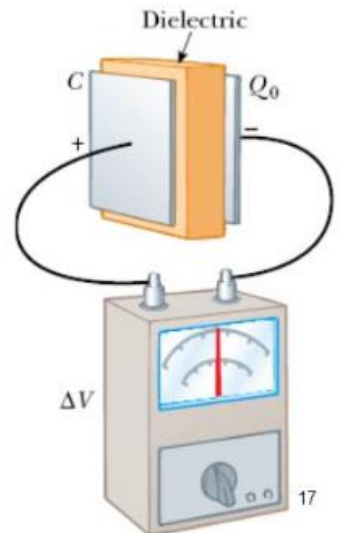
The voltages with and without the dielectric are related by the factor κ as follows:

$$\Delta V = \frac{\Delta V_0}{\kappa}$$

$$C = \kappa C_0$$

where $C_0 = \epsilon_0 A/d$

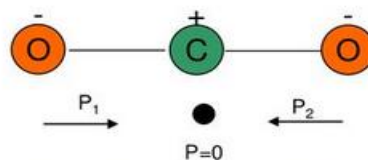
$$C = \kappa \frac{\epsilon_0 A}{d}$$



- The dielectric constant **K** varies from one material to another.
- Dielectric Materials are Insulators with no free electrons/with too low concentration of the free electrons
- When placed in electric field they are polarized.
- They are non-conductors of electricity with special properties in the presence of the electric field.
- Example: Glass, mica, polymers

NON-POLAR DIELECTRICS

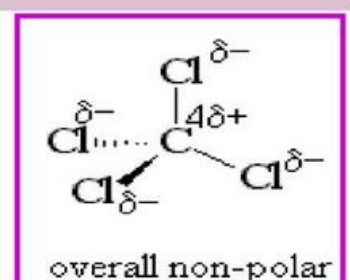
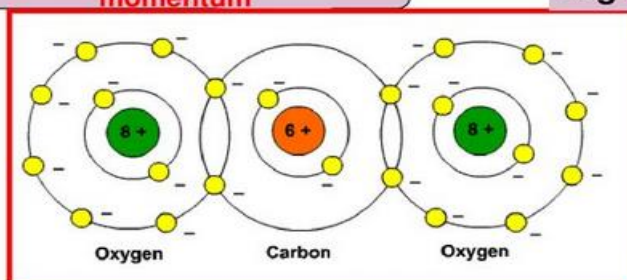
- ✓ This materials Does not posses permanent dipole moment and have centre of symmetry.
- ✓ They polarized only when placed in electric field
- ✓ Example: H_2 , N_2 , CO_2 , CCl_4



CO₂ molecule has zero dipole momentum

The polarization of non-polar molecules is independent of temperature.

They don't absorb or emit radiation of in the infrared region

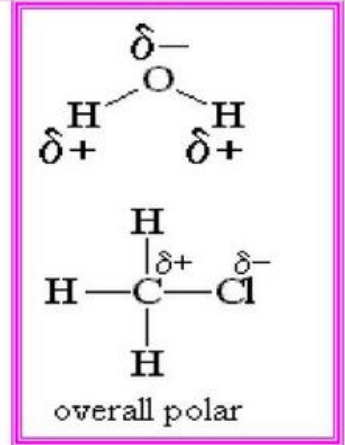
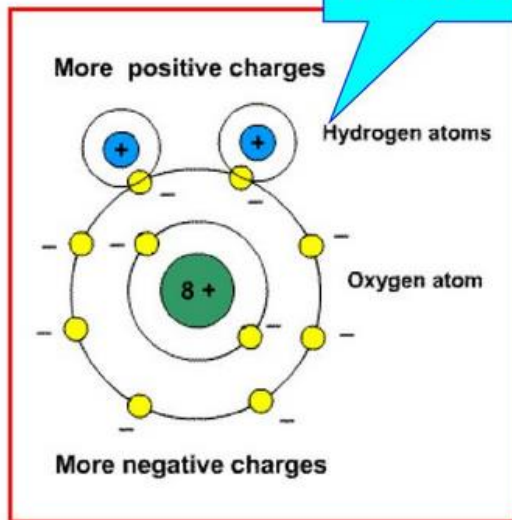


POLAR DIELECTRICS

- ✓ This material possesses permanent dipole moment even in the absence of electric field.
- ✓ Lack of centre of symmetry
- ✓ Example: HCl, H₂O

- ✓ Polarization of molecule strongly depends on the temperature.
- ✓ Molecules absorb and emit radiation in the IR region

WATER MOLECULE



DIFFERENT TYPES OF POLARIZATION IN DIELECTRICS

- ✓ The process of producing electric dipoles by an electric field is called polarization in dielectrics.
- ✓ The strength of induced dipole momentum is proportional to the applied field.

$$\mu = \alpha E$$

where α , is called as polarizability.

$$P = N\bar{\mu}$$

Polarization occurs due to several microscopic polarization mechanisms:

- ✓ Electronic Polarization (P_e)
- ✓ Ionic Polarization (P_i)
- ✓ Orientation Polarization (P_o)
- ✓ Space charge Polarization (P_s)

ELECTRONIC POLARIZATION

- ✓ It is defined as an electric strain produced in an atom by the application of electric field.
- ✓ It results from the displacement of nucleus (+Ze) and electrons (-Ze) in opposite direction in the presence of the applied field with the creation of dipole moment.
- ✓ The magnetic moment induced is proportional to the strength of field applied

$$\mu_e \propto E$$

$$\mu_e = \alpha_e E$$

where α_e is electronic polarizability of the material which is given by $\frac{\epsilon_0(\epsilon_r - 1)}{N}$
 N is number of atoms per cm^3

- ✓ Polarization is $P_e = N\alpha_e E$
- ✓ It is independent of temperature.
- ✓ Monoatomic molecules exhibits this type of polarization.

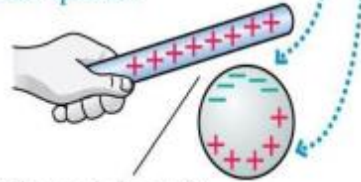
Charge polarization

- **Charge polarization** is the slight separation of the positive and negative charges in a neutral object when a charged object is brought near.

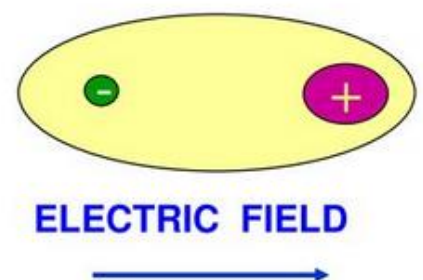
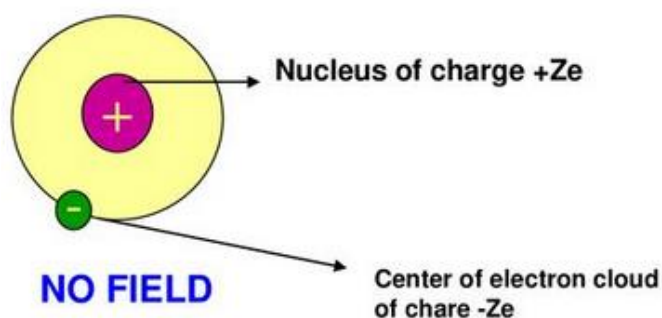
The neutral sphere contains equal amounts of positive and negative charge.



Negative charge is attracted to the positive rod. This leaves behind positive charge on the other side of the sphere.

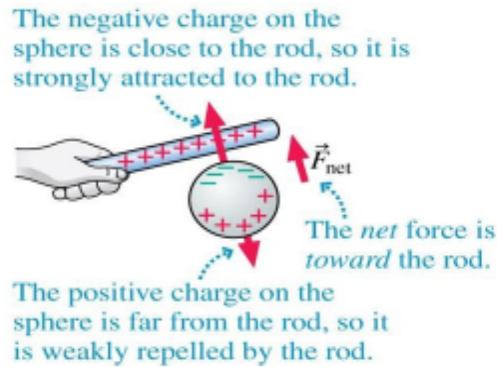


The rod doesn't touch the sphere.



Polarization

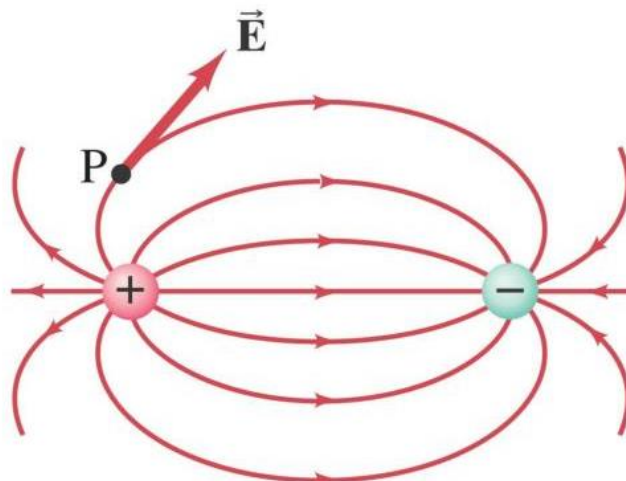
- The negative charges at the top of the sphere are more strongly attracted to the rod than the distant positive charges are repelled, so there is a net *attractive* force.
- The **polarization force** arises because the charges in a neutral object are slightly separated, *not* because the objects are oppositely charged.
- The **polarization force between a charged object and a neutral one is always attractive.**



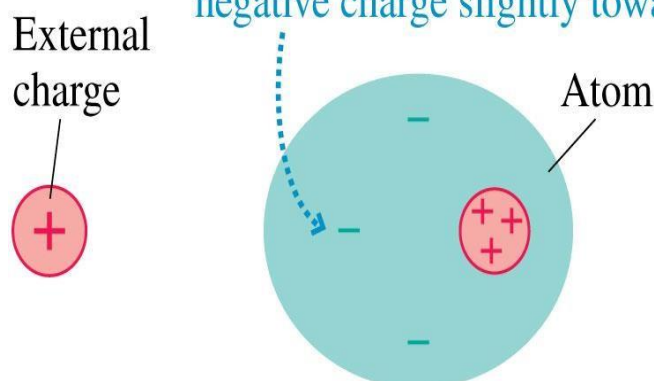
Electric Dipoles

- An **electric dipole** is two equal but opposite charges with a separation between them.

Electric dipole: two equal charges, opposite in sign:

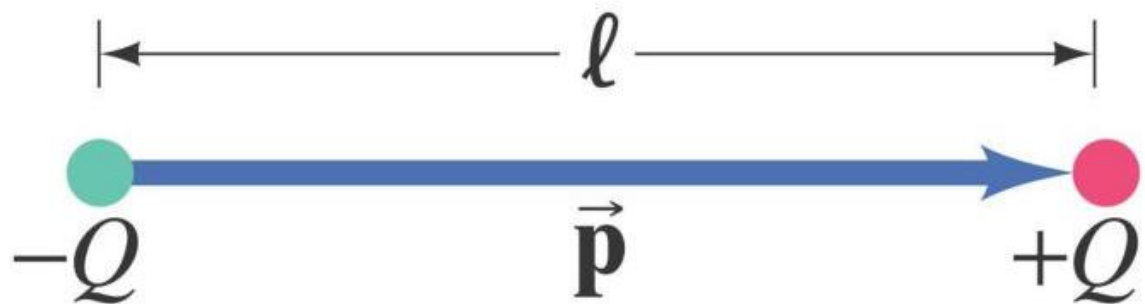


The external charge attracts the atom's negative charge, pulling the negative charge slightly toward it.

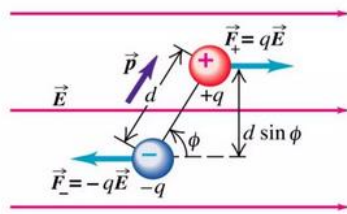


- two charged particles of magnitude q but of opposite sign, separated by a distance d . We call this configuration an *electric dipole*
- The product qd , which involves the two intrinsic properties q and d of the dipole, is the magnitude p of a vector quantity known as the **electric dipole moment** of the dipole. The direction of $\vec{p}$ is taken to be from the negative to the positive end of the dipole

An electric dipole consists of two charges Q , equal in magnitude and opposite in sign, separated by a distance ℓ . The dipole moment, $\vec{p} = Q\ell$, points from the negative to the positive charge.



Dipoles in a Uniform External Electric Field



The net force on a dipole is zero; however, there is a **non-zero torque**:

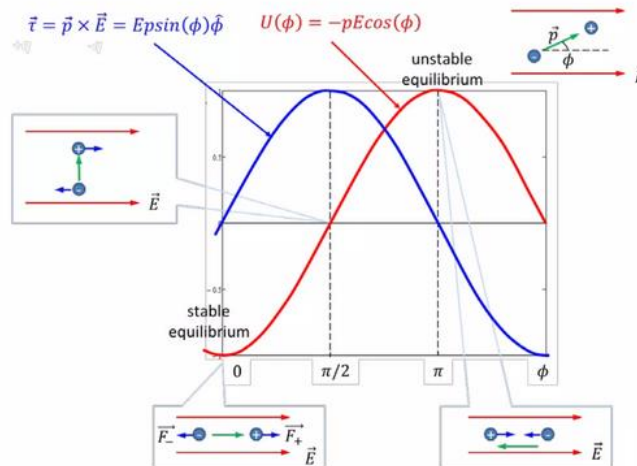
$$\tau = 2 \times Eq \frac{d}{2} \sin(\phi) = Eqd \sin(\phi) = Ep \sin(\phi)$$

p – the dipole moment

In the vector form: $\vec{\tau} = \vec{p} \times \vec{E}$ ($\vec{p}$ directed from – to +)

Potential energy of a dipole in an electric field:

$$U(\phi) = -pE \cos(\phi)$$



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Problem 1:

3 C and -3 C are two equal charges separated by a distance of 4 cm, then calculate the dipole moment of a dipole.

The dipole moment of charge 3 C and distance 4 cm will be,

$M = Q \times d$. Therefore, $M = 3 \times 0.04 = 0.12 \text{ C-m}$.

Problem 2:

An electric dipole consists of two charges, +5 μC and -5 μC , separated by a distance of 2 cm. Calculate the electric dipole moment of the system.

The electric dipole moment (p) of a system is given by the product of the charge (q) and the displacement vector (d) between the charges:

$$p = q \times d$$

Given that the magnitude of each charge (q) is $5 \mu\text{C}$ and the distance between them (d) is 2 cm (or 0.02 m):

$$q = 5 \times 10^{-6} \text{ C}$$

$$d = 0.02 \text{ m}$$

$$p = (5 \times 10^{-6} \text{ C}) \times (0.02 \text{ m})$$

$$p = 1 \times 10^{-7} \text{ C} \cdot \text{m}$$



Summary

- Electric dipole is a system in which two equal and opposite point charges are placed at a small distance.
- In the context of electricity, polarization is the process of separating opposite charges within an object.
- The electric polarization is a process of rearranging positive charges and negative charges inside an insulator which are induced by an electric field.